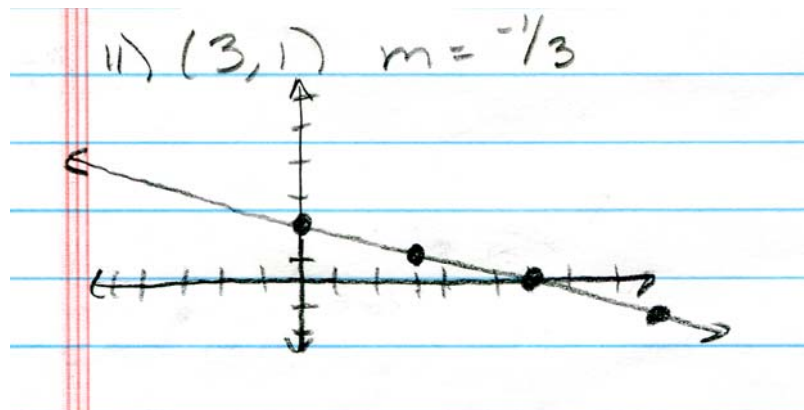


Name _____

Math 1350, Quiz #2

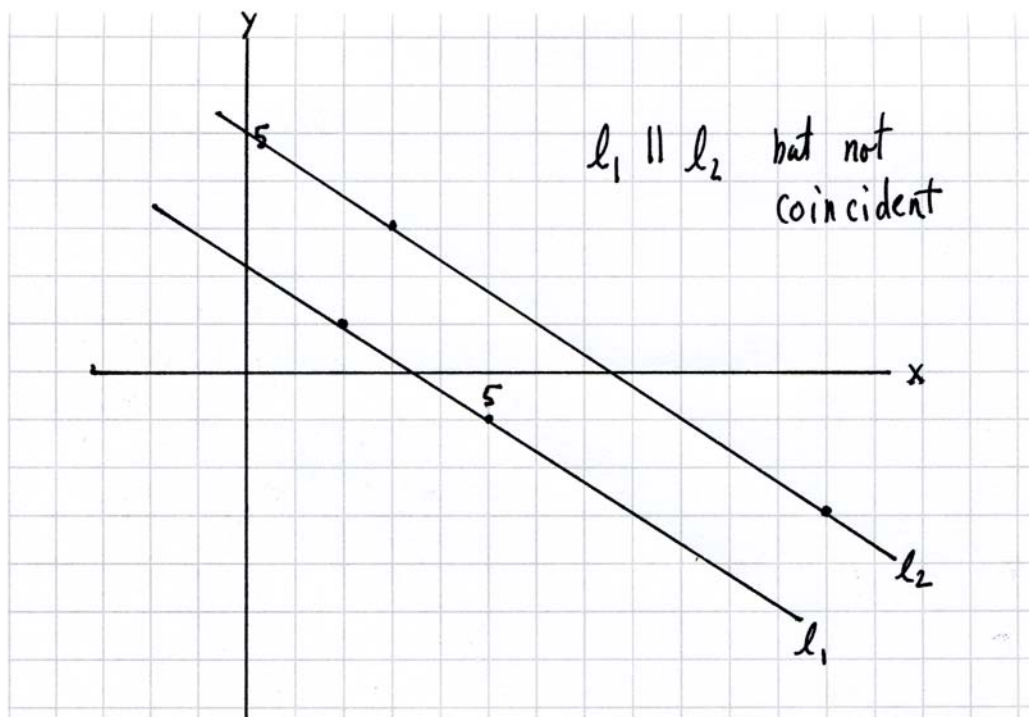
1. (Section 1.5, Problem 11) Graph the line through the point (3,1) with slope $m = -1/3$.



2. (Section 1.5, Problem 23) Find the slopes of the lines through the two pairs of points (2,1) & (5,-1) and (3,3) & (12,-3). Then, determine whether the lines are parallel, coincident, perpendicular, or none of these.

slope of line thru (2,1) & (5,-1) $m_1 = \frac{-1-1}{5-2} = -\frac{2}{3}$

slope of line thru (3,3) & (12,-3) $m_2 = \frac{-3-3}{12-3} = \frac{-6}{9} = -\frac{2}{3}$



3. (Section 1.5, Problem 29) If the line through $(x, 4)$ & $(3, 7)$ is parallel to the line through $(x, -1)$ & $(5, 1)$, find x .

$$29) (x, 4) (3, 7) \parallel (x, -1) (5, 1)$$

$$m_1 = \frac{7-4}{3-x} = \frac{3}{3-x}$$

$$m_2 = \frac{1-(-1)}{5-x} = \frac{2}{5-x}$$

// lines means slopes are equal

$$m_1 = m_2$$

$$\frac{3}{3-x} = \frac{2}{5-x}$$

$$2(3-x) = 3(5-x)$$

$$6-2x = 15-3x$$

$$6+x=15$$

$$\boxed{x=9}$$